THREE CHARACTERIZATIONS OF K-INJECTIVE COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 9869-9888, following the statement at 9848-9867.

Editorial note. No omissions: each implication is argued in full at its natural length.

Allowed standard prerequisites (editorial).

- Cohomological Hom functor. lemma-representable-homological (same chapter, used in (1) => (2)). Hom_K(-, I) is cohomological; cones of quasi-isomorphisms are acyclic.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $I^\bullet$ be a complex. The following are equivalent*

(1) *$I^\bullet$ is K-injective,*
(2) *for every quasi-isomorphism $M^\bullet \to N^\bullet$ the map*
$$\mathrm{Hom}_{K(\mathcal{A})}(N^\bullet, I^\bullet) \to \mathrm{Hom}_{K(\mathcal{A})}(M^\bullet, I^\bullet)$$
is bijective, and
(3) *for every complex $N^\bullet$ the map*
$$\mathrm{Hom}_{K(\mathcal{A})}(N^\bullet, I^\bullet) \to \mathrm{Hom}_{D(\mathcal{A})}(N^\bullet, I^\bullet)$$
is an isomorphism.

Proof. Assume (1). Then (2) holds because the functor $\mathrm{Hom}_{K(\mathcal{A})}(-, I^\bullet)$ is cohomological and the cone on a quasi-isomorphism is acyclic.

Assume (2). A morphism $N^\bullet \to I^\bullet$ in $D(\mathcal{A})$ is of the form $fs^{-1} : N^\bullet \to I^\bullet$ where $s : M^\bullet \to N^\bullet$ is a quasi-isomorphism and $f : M^\bullet \to I^\bullet$ is a map. By (2) this corresponds to a unique morphism $N^\bullet \to I^\bullet$ in $K(\mathcal{A})$, i.e., (3) holds.

Assume (3). If $M^\bullet$ is acyclic then $M^\bullet$ is isomorphic to the zero complex in $D(\mathcal{A})$ hence $\mathrm{Hom}_{D(\mathcal{A})}(M^\bullet, I^\bullet) = 0$, whence $\mathrm{Hom}_{K(\mathcal{A})}(M^\bullet, I^\bullet) = 0$ by (3), i.e., (1) holds. □

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